

# Nominal Anchors and Real Consequences: Inflation Targeting, Debt, and Investment

**preliminary**

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The views expressed herein are those of the authors and not necessarily those of the Federal Reserve Bank of San Francisco, or the Federal Reserve System.

## What we do

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secular trends in macro-finance since the 1980s often explained by real factors (demography, productivity, capital risk)

we ask: can the *monetary policy regime* itself play a role?

develop GE connections between inflation targeting and wider historical trends:

1. decline in long-term real interest rates on nominal bonds
2. increase in private (nominal) debt relative to GDP
3. investment 'drought' (declining  $I/Y$ )
4. stability of risky real returns to capital
5. strengthening of inflation targeting since the 1980s ( $\phi_\pi \uparrow$ )

can the monetary policy regime (5) be a unified explanation of (1)–(4) in a GE setting?

credible inflation targeting  $\implies$  lower inflation volatility

- nominal bonds become *safer* in real terms
- $\implies$  inflation risk premium + interest rate risk declines  $\implies R_0^* \downarrow$
- $\implies$  savers supply more savings in nominal assets  $\implies$  debt  $\uparrow$
- $\implies$  capital becomes relatively *more* risky  $\implies K \downarrow, \mathbb{E}_0[R_0^K] \uparrow$

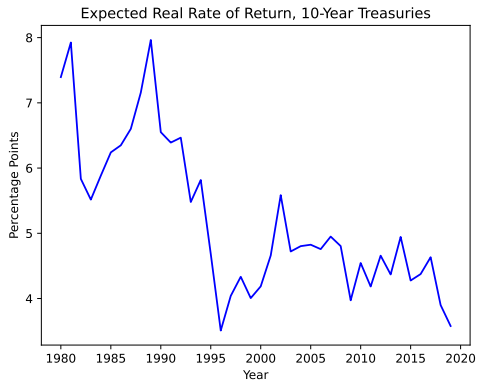
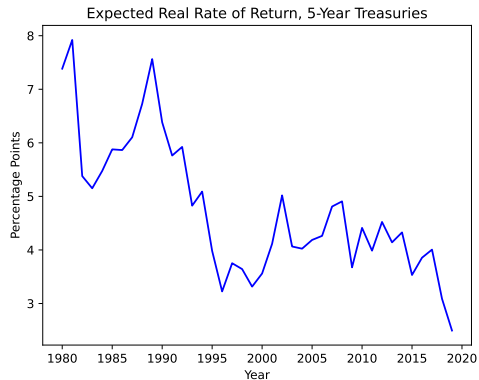
successful nominal anchoring transforms the *risk characteristics* of nominal assets, triggering portfolio reallocation away from risky capital toward safer nominal bonds

1. Stylized facts since the 1980s
2. A three-period macro-finance model
3. Analytical results: zero debt stochastic steady state
4. Analytical results: positive (nominal) debt
5. Conclusion

## Fact 1: Decline in long-term rates

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expected real yields on US nominal bonds

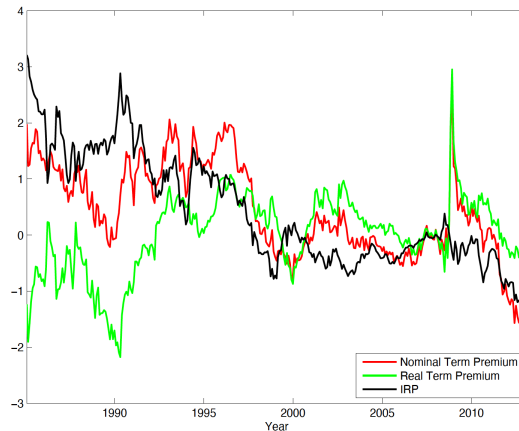


## Fact 1: Decline in long-term rates

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### Inflation Risk Premium for the UK

#### 10-year Term Premium Comparison

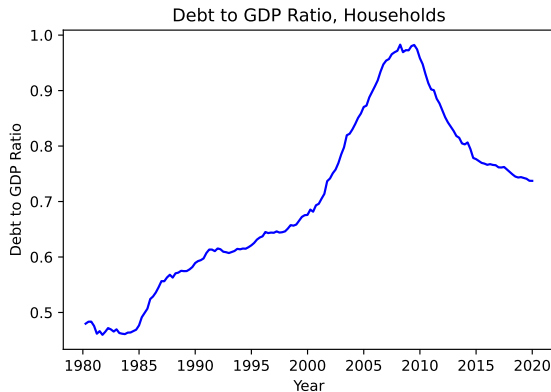


Abrahams, Adrian, Crump, and Moench (2016) for the UK.

## Fact 2: Rise in private (nominal) debt

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household debt relative to GDP in the US increased since 1980

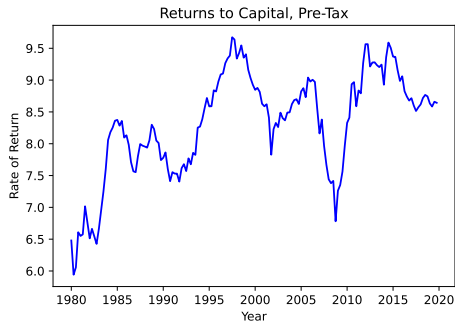
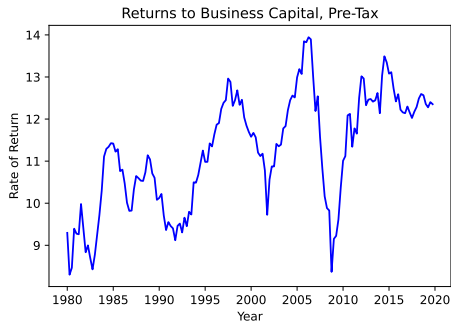


Source: Our calculations based on Mian Straub & Sufi (2023)

### Fact 3: Returns to capital stable or rising

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real return to capital, US data

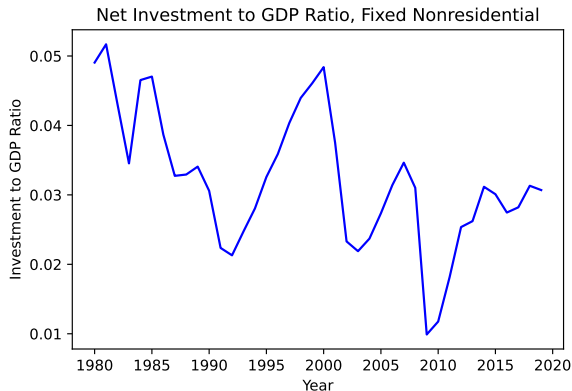


Source: based on Gomme, Ravikumar, & Rupert (2019)

## Fact 4: Investment drought

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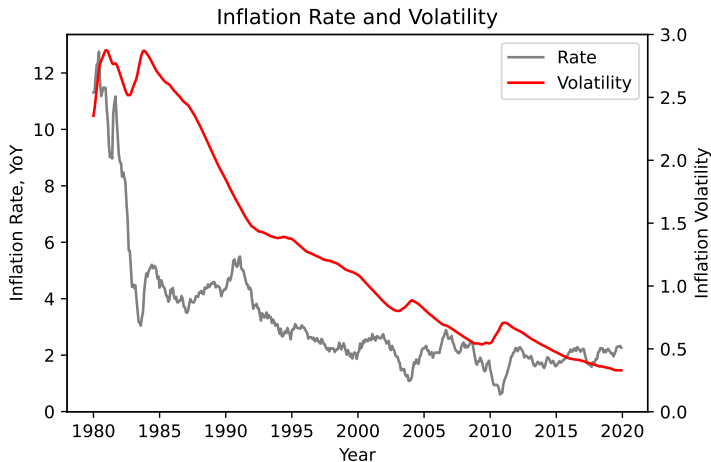
investment to GDP declined



Source: our calculations

## Fact 5: Strengthening of inflation targeting

post Volcker, central banks adopt inflation targeting. US data ( $\pi$ , level and rolling sd)



**Three Periods:** ( $t = 0, 1, 2$ ), flexible price economy

### Agents

- Borrower and savers
- Central bank (Taylor rule)

### Assets

- Short-term nominal bond (zero supply)
- Long-term nominal bond
- Capital: illiquid, cannot sell at date 1

### Endowments

- exogenous endowment at all dates

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### Shocks

All uncertainty resolves at date 1

### Aggregate Risk

- Preference shock  $\xi_1$
- Source of inflation volatility
- inflation risk premium

### Idiosyncratic Liquidity Risk on Savers

- Prob.  $\lambda$ : forced to liquidate at date 1
- Low income  $y_L$  in liquidity state
- interest rate risk on long-term bond

**Two types**  $j \in \{b, s\}$ : borrowers and savers, each mass one

- Borrowers: discount factor  $\beta_b^t$ , with  $\beta_b$  small (always constrained)
- Savers: discount factor  $\beta^t \xi_t$ , where  $\xi_t$  is a stochastic preference shock

**Preference shock**  $\xi_1$ :

- $\xi_0 = 1$ ,  $\xi_1 \sim [\xi_L, \xi_H]$  with  $\mathbb{E}[\xi_1] = 1$ ,  $\text{Var}(\xi_1) > 0$
- Date-2 persistence:  $\xi_2 = \xi_1^\rho$  with  $\rho \in (0, 1)$
- All uncertainty resolves at date 1

## Idiosyncratic Liquidity Shock

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Each saver draws  $\eta \in \{0, 1\}$  (wt on date 2 utility) at beginning of date 1, i.i.d., independent of  $\xi_1$ :

|                      | Impatient ( $\eta = 0$ ) | Patient ( $\eta = 1$ ) |
|----------------------|--------------------------|------------------------|
| Probability          | $\lambda$                | $1 - \lambda$          |
| Date-1 endowment     | $y_L$                    | $y_H \geq y_L$         |
| Long bonds at date 1 | Sells at $Q_1^L$         | Holds to maturity      |

**Saver's date-0 utility:**

$$U_s = u(C_{s,0}) + \beta \mathbb{E} \left[ \xi_1 \left( \lambda u(C_{s,1}^I) + (1 - \lambda) \left[ u(C_{s,1}^P) + \beta \xi_1^{p-1} u(C_{s,2}^P) \right] \right) \right]$$

$$u(C) = C^{1-\gamma}/(1-\gamma)$$

1. **Short-term nominal bond** — issued at date 0, pays  $(1 + i_0^S)$  at date 1
  - Zero supply: pins short-term real interest rate
  - Real return:  $(1 + i_0^S)/\Pi_1$  — exposed to date-1 inflation
2. **Long-term nominal bond** — issued at date 0 at price  $Q_0^L$ , pays 1 at date 2
  - Liquid: can be resold at date 1 at price  $Q_1^L = \frac{1}{1+i_1^S}$
  - Impatient type sells at  $Q_1^L \Rightarrow$  exposed to both inflation risk and interest rate risk
3. **Real capital  $K$**  — invested at date 0, produces  $\bar{A}K^\alpha$  at date 2
  - Illiquid: cannot be sold or pledged at date 1
  - No inflation exposure

## Saver's Budget Constraints

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**Date 0** (before learning  $\eta$ ):

$$P_0 C_{s,0} + P_0 K + Q_0^L B_s^L = P_0 Y_0$$

**Date 1, Impatient type** ( $\lambda$ ): sells all long bonds, consumes everything

$$P_1 C_{s,1}^I = P_1 y_L + Q_1^L B_s^L$$

**Date 1, Patient type** ( $1 - \lambda$ ): receives  $y_H$ , holds bonds to maturity, saves  $s_1$

$$P_1 C_{s,1}^P + \frac{\lambda}{1 - \lambda} Q_1^L B_s^L = P_1 y_H$$

$$P_2 C_{s,2}^P = P_2 \bar{A} K^\alpha + P_2 Y_2 + \frac{B_s^L}{(1 - \lambda)}$$

Issue long-term nominal bonds, face debt limit (on expected repayment)

$$\mathbb{E}_0 \left[ \frac{B_b^L}{P_2} \right] \geq -d^L.$$

- Consume at dates 0 and 2 only:

$$P_0 C_{b,0} = P_0 Y_{b,0} + Q_0^L B_b^L, \quad P_2 C_{b,2} = P_2 Y_{b,2} - B_b^L$$

- Always constrained ( $\beta_b$  small)

### Taylor rule:

$$1 + i_1^S = \frac{1}{\beta} \Pi_1^{\phi_\pi}, \quad \phi_\pi > 1$$

- policy regime characterized by  $\phi_\pi > 1$
- normalize  $P_0 = 1$
- Assume  $\Pi_2 = 1$  (Krugman 1998)

### Date-1 long bond price:

$$Q_1^L = \frac{1}{1 + i_1^S} = \beta \Pi_1^{-\phi_\pi}$$

⇒ Links monetary policy to liquidity risk:  $Q_1^L$  fluctuates with  $\xi_1$  through  $\Pi_1$

⇒ More aggressive targeting stabilizes the resale price of long bonds

### Three pricing equations:

- **Short bond** (no supply):

$$1 = \mathbb{E} \left[ (\lambda M'_{01} + (1 - \lambda) M^P_{01}) \frac{1 + i_0^S}{\Pi_1} \right]$$

- **Long bond** (hybrid: impatient sells at date 1, patient holds to date 2):

$$Q_0^L = \mathbb{E} \left[ \lambda M'_{01} \frac{Q_1^L}{\Pi_1} + (1 - \lambda) M^P_{02} \frac{1}{\Pi_1 \Pi_2} \right]$$

- **Capital** (only patient types benefit, liquidity tax  $(1 - \lambda)$ ):

$$1 = (1 - \lambda) \mathbb{E} [M^P_{02} R_K], \quad R_K = \alpha \bar{A} K^{\alpha-1}$$

allocations, prices, interest rates such that

1. households optimize,
2. bond markets clear, and
3. the Taylor rule holds

## Zero-debt benchmark

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Set  $d_0 = 0$ : no borrowing or lending. savers only save in capital.

from patient type's date-1 Euler + Taylor rule, solve for equilibrium inflation:

$$\Pi_1 = \bar{D} \xi_1^a, \quad \text{where } a \equiv \frac{1 - \rho}{\phi_\pi}, \quad \bar{D} \equiv \left( \frac{Y_2 + \bar{A}\bar{K}^\alpha}{y_H} \right)^{\gamma/\phi_\pi}$$

inflation variance:

$$\text{Var}(\log \Pi_1) = \left( \frac{1 - \rho}{\phi_\pi} \right)^2 \text{Var}(\log \xi_1) \quad \implies \quad \text{strictly decreasing in } \phi_\pi$$

cumulative price level:  $P_2 = \Pi_1 \Pi_2 = \Pi_1$

## Zero-debt: short-bond inflation risk premium

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expected real return on the short nominal bond:

$$\bar{R}_0^{S*} = \bar{R}_1^f \cdot \underbrace{\frac{\mathbb{E}[\xi_1^{-a}]}{\mathbb{E}[\xi_1^{1-a}]}}_{\text{IRP}^S > 1}$$

- $\uparrow \phi_\pi \implies \downarrow \bar{R}_0^{S*}$  (lower compensation for inflation risk)
- as  $\phi_\pi \rightarrow \infty$ :  $\text{IRP}^S \rightarrow 1$ . Perfect IT eliminates inflation risk premium
- $\text{IRP}^S$  independent of  $\gamma$ ,  $\lambda$ , or  $\omega$

## Decomposing the real bond yield

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use bond pricing equation  $1 = \mathbb{E}[M \cdot (1 + i_0^S)/\Pi_1]$ :

$$R_0^{S*} = \underbrace{\frac{1}{\mathbb{E}[M_{01}]}_{R^f} + \underbrace{\left( -\frac{\text{Cov}(M_{01}, (1 + i_0^S)/\Pi_1)}{\mathbb{E}[M_{01}]} \right)}_{\text{inflation risk premium } \psi}$$

at  $b^L = 0$ :  $C_{s,1}$  is deterministic for each type  $\implies R^f = \bar{R}_1^f$  is invariant to  $\phi_\pi$

$$\uparrow \phi_\pi \implies \downarrow \psi \implies \downarrow R_0^{S*}$$

## Why does stronger targeting lower the inflation risk premium?

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At zero debt,

$$\log M_{01} = \text{const.} + \log \xi_1, \quad \log \Pi_1^{-1} = \text{const.} - a \log \xi_1, \quad a = \frac{1 - \rho}{\phi_\pi}.$$

**State alignment is unchanged**

$$\text{Corr}(\log M_{01}, \log \Pi_1^{-1}) = -1.$$

**But exposure shrinks**

$$\text{sd}(\log \Pi_1^{-1}) = a \text{sd}(\log \xi_1),$$

Nominal bonds still pay poorly when  
marginal utility is high.

and

$$\phi_\pi \uparrow \implies a \downarrow.$$

$$\phi_\pi \uparrow \implies \text{nominal bonds become safer in real terms} \implies \text{IRP}^S \downarrow$$

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**safe asset demand** story:  $\uparrow \phi_\pi$  reduces volatility of real bond payoffs  $\implies \bar{R}_0^{S*} \downarrow$   
bonds remain bad hedges

## Zero-debt: capital is independent of $\phi_\pi$

---

capital Euler (only patient types benefit; liquidity tax  $1 - \lambda$ ):

$$1 = (1 - \lambda) \mathbb{E} \left[ M_{02}^P R_K \right], \quad R_K = \alpha \bar{A} K^{\alpha-1}$$

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at  $b^L = 0$ :  $M_{02}^P = \beta^2 \xi_1^\rho \left( \frac{Y_2 + \bar{A} K^\alpha}{C_{s,0}} \right)^{-\gamma}$

- $R_K$  is independent of  $\phi_\pi$ : capital has no inflation exposure
- $R_K$  is increasing in  $\lambda$ : liquidity tax requires higher compensation
- $R_K$  is independent of  $\omega = \left( \frac{y_H}{y_L} \right)^\gamma \geq 1$ : capital priced by patient type only

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**Illiquidity premium:**  $\sqrt{R_K} - \bar{R}_0^{L*}$

- $R_K$  invariant to  $\phi_\pi$ , while  $\bar{R}_0^{L*}$  strictly decreasing
- $\implies$  **spread widens** as inflation targeting becomes more aggressive
- $R_K$  independent of  $\omega$ , but bond price increasing in  $\omega \implies$  income risk amplifies

**At zero debt**, more aggressive inflation targeting ( $\uparrow \phi_\pi$ ):

1.  $\bar{R}_0^{S*} \downarrow$ : short-bond IRP falls (inflation volatility channel, 100%)
2.  $\bar{R}_0^{L*} \downarrow$ : long-bond IRP falls (same mechanism)
3.  $R_K$  unchanged: capital return pinned by real fundamentals
4. illiquidity premium  $\sqrt{\bar{R}_K} - \bar{R}_0^{L*}$  widens

Next: what happens when we turn on debt ( $b^L > 0$ )?

## Positive debt: perturbation around stochastic steady state

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expand around the **stochastic** zero-debt equilibrium (not a deterministic SS)

- inflation is stochastic even at  $b^L = 0$
- $\implies$  nonzero inflation risk premium embedded from the outset
- contrasts with standard perturbation where risk premia vanish at first order

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at  $b^L > 0$ : impatient type's consumption becomes **stochastic**

$$C'_{s,1} = y_L + \frac{Q_1^L}{P_1} B_s^L = y_L + \beta \Pi_1^{-(1+\phi_\pi)} \frac{b^L}{Q_0^L}$$

- $C'_{s,1}$  depends on  $\Pi_1$  through liquidation payoff  $\propto \Pi_1^{-(1+\phi_\pi)}$
- low base  $y_L \implies$  marginal utility amplified by  $\omega = (y_H/y_L)^\gamma$
- $M'_{01}$  now covaries with inflation  $\implies$  interest rate risk channel activated

**Proposition.** At  $b^L > 0$ ,  $\partial \bar{R}_0^{L*} / \partial \phi_\pi < 0$  through two channels:

**(i) Inflation risk channel** (present for all  $\lambda$ ):

- $\uparrow \phi_\pi$  reduces  $\text{Var}(\log \Pi_1) \implies$  compresses IRP
- same mechanism as zero-debt benchmark

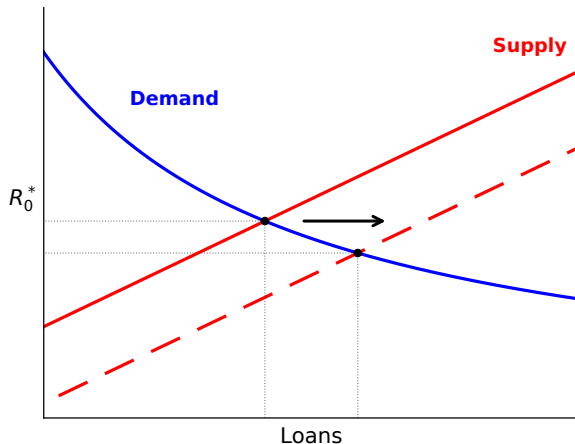
**(ii) Interest rate risk channel** ( $\propto \lambda$ , amplified by  $\omega$ ):

- $\uparrow \phi_\pi$  stabilizes  $Q_1^L = \beta \Pi_1^{-\phi_\pi} \implies$  reduces capital-gain risk
- priced by impatient type's elevated SDF ( $\omega$  times patient type's)
- vanishes at  $\lambda = 0$ ; strengthens in  $\lambda$  and  $\omega$

## Intuition: supply and demand for credit

more aggressive inflation targeting  $\implies$  lower  $R_0^{L*}$

$\implies$  more nominal debt supplied by savers  $\implies$  lower  $R_0^L$ , more credit in equilibrium



[binding debt limit means the demand curve is  $R_0^{L*} = d_0/\text{Loans}$ ]

## Crowd-out: the question

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when bonds become safer, do savers **displace capital** or just **save more**?

write total savings as  $S_0 \equiv Y_0 - C_{s,0} = K + b^L$

differentiate with respect to bond holdings  $b$ :

$$\left. \frac{dK}{db} \right|_{b=0} = \underbrace{-1}_{\text{portfolio margin}} + \underbrace{\left. \frac{dS_0}{db} \right|_{b=0}}_{\text{savings margin}}$$

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- **portfolio margin** =  $-1$ : holding  $S_0$  fixed, each dollar of bonds displaces a dollar of capital
- **savings margin** =  $1 + k$ : when bonds are more attractive, saver may cut  $C_{s,0}$  and save more
- $k \equiv dK/db$  is the **crowd-out coefficient**

## Crowd-out: three forces

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linearize capital Euler  $(1 - \lambda) \mathbb{E}[M_{02}^P R_K] = 1$  around  $b = 0$  (stochastic):

$$0 = \underbrace{(1+k) \frac{\gamma}{\bar{C}_0}}_{\text{(a) date-0 smoothing}} + \underbrace{k \frac{1-\alpha}{\bar{K}}}_{\text{(b) diminishing returns}} + \underbrace{\left(k + \frac{1}{\chi}\right) \frac{\gamma R_K}{\bar{C}_2}}_{\text{(c) date-2 smoothing}}, \quad \chi \equiv 1 + \lambda(\omega - 1).$$

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### (a) Date-0 consumption smoothing

- saver saves more ( $1 + k > 0$ )  $\implies$  cuts  $C_{s,0} \implies$  higher marginal utility today
- resists further saving  $\implies$  limits the savings margin

### (b) Diminishing returns (absorbs displacement)

- capital falls ( $k < 0$ )  $\implies$  marginal product  $R_K$  rises  $\implies$  partially stabilizes capital

### (c) Date-2 consumption smoothing (limits crowd-out)

- capital falls  $\implies \bar{C}_2 = \bar{A}K^\alpha + Y_2$  falls  $\implies$  higher marginal utility at date 2
- makes remaining capital more valuable  $\implies$  resists further displacement

## Capital extension: implications

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$$\uparrow \phi_{\pi} \implies \text{safer nominal bonds} \implies b \uparrow \implies K \downarrow \implies \mathbb{E}_0[R_0^K] \uparrow$$

mechanism: improved risk-return profile of nominal bonds crowds out capital investment

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### Equilibrium implications ( $\uparrow \phi_\pi$ ):

- $\bar{R}_0^{L*} \downarrow$  (declining real rate on nominal bonds)
- debt  $\uparrow$  (rising private credit)
- $K \downarrow$  (investment drought)
- $R_K$  stable (pinned by real fundamentals)
- $\sqrt{R_K} - \bar{R}_0^{L*} \uparrow$  (widening safe-risky spread)

aggressive inflation targeting regime:

- reduces inflation volatility  $\implies$  lowers  $R_0^*$
- low inflation vol & interest rate risk  $\implies$  nominal bonds safer  $\implies$  supply of savings in nominal assets goes up
- capital becomes relatively *more* risky  $\implies$  supply of savings in real assets falls
- equilibrium:  $\uparrow$  private credit,  $\downarrow$  natural rate on nominal bonds,  $\downarrow$  investment

a unified account of macro-finance trends since the 1980s:

declining real rates, rising debt, investment drought, stable returns on capital  
all as equilibrium responses to a single shift in monetary policy regime

## What next

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quantitative exercise:

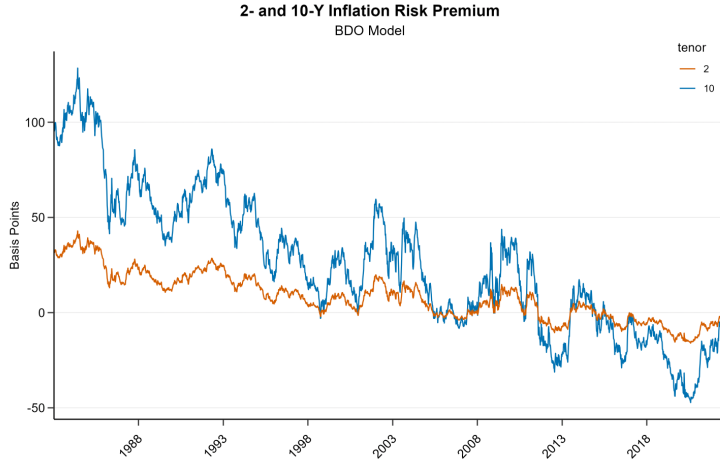
- infinite horizon model with aggregate uncertainty
- decompose the contribution to decline in  $r^*$  from  $\phi_\pi$  vs other forces (demography, productivity, capital risk)

THANK YOU!

## Fact 1: Decline in long-term rates

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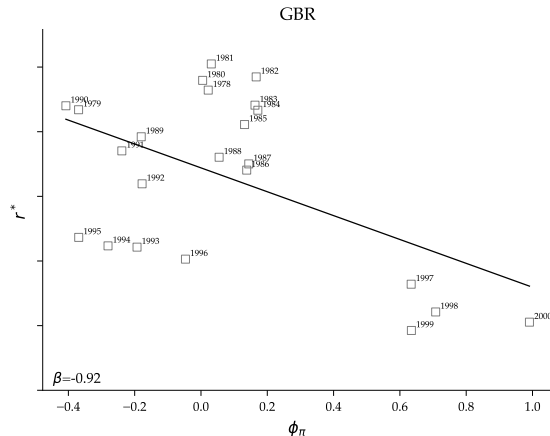
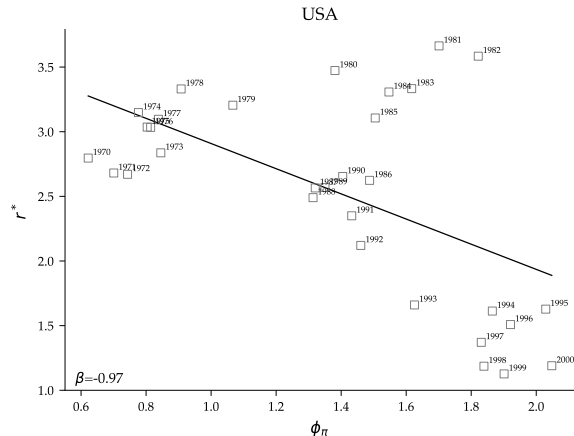
### Inflation Risk Premium for the US



Breach, D'Amico, Orphanides (2020) for the US.

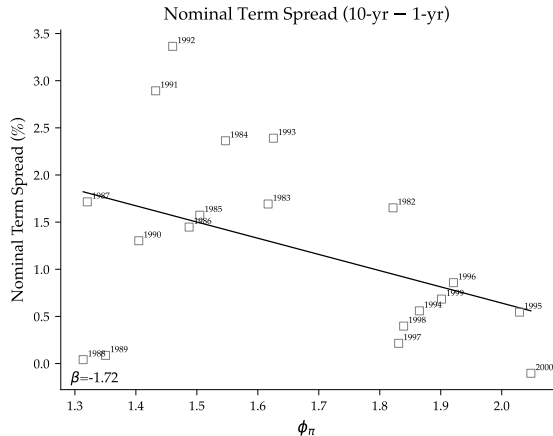
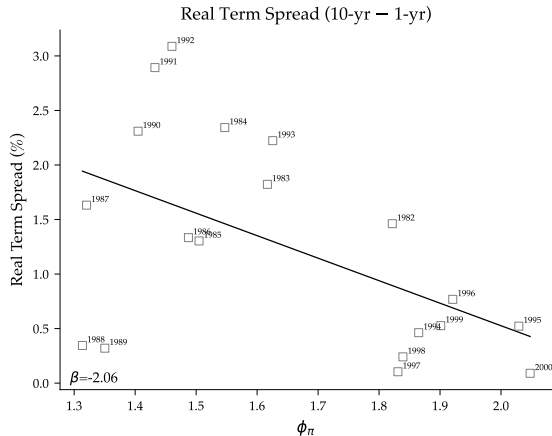
# Monetary policy and r-star

HLW estimates of  $r^*$  and our estimates of  $\phi_\pi$  based on Carvalho Nechio Tristao (2021)



# Monetary policy and term spread

Term spread (10 y - 1 y) + our estimates of  $\phi_\pi$  based on Carvalho Nechio Tristao (2021)



# Monetary policy and DGGT r-star spread

DGGT term spread + our estimates of  $\phi_\pi$  based on Carvalho Nechio Tristao (2021)

